CS-001 Part, Mold & Material Labeling

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Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer		
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.
C	2026-08-03	Claude	Out of outline. §7 written in full: the two keep-out rules, where each class of object gets its label, the three mold identification layers, pair-print relocation for parts, and the label content per class. §5 no longer claims nothing needs ordering approval, because a label printer does (CS-012). §4 defines the ID and the QR. §8 gains checks that can actually be walked.

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1. Purpose

Unlabeled molds sat blocking shared storage for weeks while nobody was sure whose they were (PP-10), and unlabeled offcuts made “wait, which gsm is this?” a weekly guessing game (PP-09). So: everything physical gets a label that points back to a record.

Rev C exists because Rev B said that and then stopped. It named no media, so labels were Sharpie on masking tape, which the shop's own isopropyl wipe dissolves (R7). It gave no keep-out rule, so a label on a flange is a vacuum leak nobody can find at 27 inHg. And it predates the app being able to print a label at all, which it now can.

2. Scope

Molds, parts (from demold onward), tooling-board stock, cloth rolls and offcuts, resin and hardener containers, mixed-chemical containers, test panels and coupons, jigs and fixtures, and storage locations.

Not covered: electrical and wiring labeling, which is electrical's own standard, and anything not physically ours.

3. References

Ref	Document	Where
R1	PP-09, PP-10	FEB-COMP-PP-001
R2	CS-013 (the record a label points to, and the ID grammar)	this series
R3	CS-011 (home locations, the storage map)	this series
R4	CS-003 §7.3 (glue-up), CS-005 (machining), CS-004 §7 (sealing and release)	this series
R5	CS-009 (trim, sand, bond, clear coat)	this series
R6	CS-012 (the label printer is a purchase)	this series
R7	CS-004 §5 and §7.1: "Never acetone, never fibrous shop towels"; IPA 90% wipe at every stage	CS-004
R8	ISO/IEC 18004 (QR symbology: alphanumeric mode, 4-module quiet zone)	published standard
R9	ASTM D3039 (tensile coupon geometry: gauge and grip sections)	published standard

4. Definitions

- **ID** — the record's own identifier, `PREFIX-SNx-###`, assigned by the app and never reused (CS-013 §4). The ID *is* the label text; there is no separate part number.
- **Label** — a printed 4 × 1 inch card carrying, at minimum, the **ID**, the **class word**, the **name**, and the **one fact that identifies that class of object** (§7.6). Most also carry a QR code.
- **Class word** — PART, MOLD, TEST PANEL, FABRIC, RESIN, BOARD, JIG, STORAGE. Mandatory, and larger on the label than the code itself, so `P-SN6-007` can never be read as `PROJ-SN6-007` at 7 pt with resin on it.
- **QR code** — encodes `HTTPS://FEB-COMPOSITES.WEB.APP/Q/<ID>`. Uppercase, no query string, no fragment. Scanning it with any phone camera opens a public page naming the object, needing no account and no signal.
- **Mark** — a hand-written identifier in oil-based paint marker, used where a label cannot go (§7.4, §7.7).
- **Engraving** — the ID cut into a mold during machining. The record of record (§7.4).

The QR payload must stay inside QR alphanumeric mode (`0-9 A-Z space $%*+-./:`, per R8). One lowercase letter, one `?utm=`, or a `#` fragment pushes the code from version 3 to version 4 and drops its error correction from 25% to 15%, at the same physical size. The printed label looks identical and simply scans worse once it has resin on it.

5. Equipment & Materials

Item	Spec / product	Source	Notes
Paint marker	Oil-based paint marker, fine, white or silver	Campus store, ~$5	**Replaces the Sharpie for every hand mark.** A standard Sharpie is IPA-soluble, and the process wipes with 90% IPA at every stage (R7).
Label stock, everyday	Avery 5161, 1 × 4 in, 20 per sheet	Any office supplier	Prints on the existing B&W laser.
Label stock, chemicals	Avery 5522 WeatherProof polyester, 1 × 4 in, 14 per sheet	Any office supplier	Survives a hardener drip; paper does not. The extra third of an inch carries the hazard strip.
Overlaminate	3 in clear packing tape	Already in the shop	§7.8. Must be **wider than the label**.
Label printer	Brother PT-D610BT with TZe-251 24 mm laminated tape	~$115, tape ~$19/roll	**Needs ordering approval under CS-012.** Not required to start: §7.8 is the documented zero-cost path and stays the fallback.
Tags	Laminating pouches and 100 mm zip ties	Campus store	For anything tag-able: rolls, remnants, bins, coupon sets.

Rev B's §5 said "Nothing that needs ordering approval." That stopped being true the moment a label printer entered the plan. The printer goes through CS-012 like any other purchase; nothing else on this list does.

6. Safety

Labeling has no physical hazards of its own. Three rules exist to stop labeling from *creating* hazards or scrap:

1. **Never label a mold's working surface.** Anything on the tool face telegraphs through into the part; a 0.08 mm label edge prints through two plies.
2. **Never label within 40 mm of a flange perimeter, on either face.** Tacky tape is 12.7 mm wide, and a label edge under or across the seal is a leak channel you will not find at 27 inHg. CS-006 §7.4.3 already names tacky-tape seams as the first place to hunt leaks; do not add a second cause.
3. **No silicone-adhesive tape anywhere in the shop**, which is Kapton and most amber films. Trace silicone causes fisheyes in clear coat and kills DP420 adhesion. Its heat resistance buys nothing here: there is no oven and no post-cure (CS-006 §5).

7. Procedure

7.1 Printing a label

1. Open the record in the app. Press **Label** on a work order, part, mold, material lot or item, or use **Labels** under Reports for a batch.
2. Choose the stock: Avery 5161 for everyday, 5522 WeatherProof for anything that will get resin or hardener on it.
3. If the sheet has already been partly used, set **Start at cell** to the first unused one. A three-label batch onto a fresh sheet wastes seventeen.
4. Leave the calibration bar on. Print, then **measure it with a steel rule: it must be exactly 100 mm.** If it is not, the browser applied “Fit to page” scaling and every label on the sheet is misregistered. Set printer scaling to 100% or Actual Size and reprint.
5. Peel and apply per §7.2.

   Browsers apply that scaling silently. Ten seconds with a rule is cheaper than a wasted sheet of polyester.

7.2 Where each class of object gets its label

Object	Where it goes	Media
Mold	**Back face**, at least 25 mm inboard of the outline on every side. Never the working surface, never the flange (§6).	Engraved plus a printed label (§7.4)
Part, at demold	Trim area, bag side, inside the CAD trim line	Printed, permanent adhesive
Part, after trim	A hidden face, 25 mm from any DP420 bond line, never a face that gets bagged for a secondary bond	Printed, applied before clear coat (§7.5)
Test panel	Bag side, in the offcut margin, **plus** coupon IDs marked across the cut lines before sawing (§7.6)	Printed plus paint marker
Tensile coupon	**Paint marker on the tab only**, outside the gauge section. No adhesive label.	Marker, or a pouch tag on a zip tie through the tab waste
Fabric roll	A tag through the core, on wire or a zip tie	Printed label on the tag
Fabric offcut bag	On the bag, with a blank field for the remaining length	Printed, hand-write the quantity
Resin, hardener	On the container	**Polyester only** (5522 or TZe)
Mixed pot	Write-on tape and marker (§7.7). Nothing printed, ever.	Tape
Tooling board sheet	Direct paint marker on the face, 15 mm characters	Marker
Board remnant	Marker for the ID, plus a taped tag for dimensions	Marker plus tag
Storage location	Front edge of the shelf, readable standing up	Printed
Jig, fixture	A non-clamping face	Printed; stamped aluminium if it outlives the season

A coupon label in the gauge section changes the cross-section and can seed the failure; in the grip section it changes thickness and causes slip (R9). Marker on the tab is the only correct answer.

7.3 The general rule

The durable answer for a **tool** is subtractive: engrave it. For a **part** it is encapsulation: put the label under the clear coat. For **everything else** it is laminated media. Choose in that order.

7.4 A mold is identified three times

There is no moment when a flange label is clean: XCR goes over the whole working surface and the flanges, then four coats of wax, then PVA. So a mold gets three artifacts with three lifetimes.

1. **At glue-up (CS-003 §7.3).** Mark the part name in paint marker, 15 mm characters, on **two adjacent side faces** of the glued stack. Between glue-up and the CNC slot a glued block sits on a shelf beside three other glued blocks, which is the highest-confusion and lowest-cost-to-fix moment in the whole process. Mark faces the toolpath will remove less than 6 mm from, or mark the bottom.
2. **At machining (CS-005).** Engrave the mold ID on the back face: 6 mm cap height, 1.0 mm deep, single-line font, 3.175 mm ball or 90° V-bit. The machine is already set up and the part already fixtured, so it costs about 90 seconds. **This is the record of record.** It survives IPA, acetone, four coats of wax, PVA, sanding, the label falling off, and the printer being broken. Every other layer is a convenience wrapped around it.
3. **At the end of sealing (CS-004 §7, step 4 "Identify").** After the fourth wax coat is buffed and the PVA is dry, so every wet operation is finished. Scuff a patch of the back face to 220 grit, wipe with IPA, apply the printed label. Tooling board is porous; without the scuff the adhesive has nothing to key into and the label curls off within a month.

   Do not apply the printed label before sealing. It either gets brushed over with XCR, becoming a permanent bump in the seal coat, or it gets peeled off and leaves adhesive that XCR will not wet out.

Re-sealing. When a mold is stripped and re-sealed, the old label comes off with the strip and a **new one is printed**, carrying the incremented parts-pulled count and the new seal date. That is a feature: a label reading `SEALED 2025-11-02 · USES 09` is telling you something important, and forcing the reprint is how it stays true.

7.5 A part's label moves once, by printing two

Do not peel and re-stick. A label peeled off a demolded part carries wax and PVA on its adhesive. It will not restick to a freshly sanded face, and if it does, it puts release agent onto a surface about to be clear-coated.

1. **At demold (CS-006 / CS-007).** Print **two identical labels**. Apply label A to the trim area on the bag side, inside the CAD trim line. Staple label B to the work order traveler.
2. **At trim (CS-009).** Label A is cut off and discarded with the trim. Identity now lives on the traveler, which is physically with the part.
3. **After sanding, before clear coat (CS-009).** Apply label B to the scuffed, IPA-wiped hidden face.
4. **Clear coat (CS-009).** The acrylic urethane goes over label B where geometry allows, encapsulating it permanently.

7.6 What the label says

Every label carries the ID, the class word and the name. The line under those is **the one fact that identifies that class of object**, and it is the reason the label is worth printing:

Class	The identifying fact
Part, test panel	**The layup stack**, in CS-002 shorthand (`6X 195 TWILL + .125 NOMEX`)
Mold	**The sealing record**: sealed date, by whom, parts pulled, revision

Class	The identifying fact
Fabric, resin, consumable	**The vendor lot number**, with opened and expiry dates
Board, remnant	Dimensions and density
Storage location	Nothing further; the name is the fact

"What layup stack was used for seat last year" is the question PP-09 records nobody being able to answer. It is on the label, so it can be answered with the phone still in your pocket.

Coupons. Write every coupon ID on the panel **before the first cut**, across the cut lines, so each piece inherits a legible fragment. A coupon is `PNL-SNx-###-Cnn`, where `-Cnn` is the cut sequence left to right. Coupon labels are text only: 12 mm tape has 8 mm of print height, which cannot hold even a version 1 QR code with its quiet zone.

7.7 Mixed chemicals

A pot of XCR has an 8 minute pot life at 20 °C (CS-004 §7.3). Printing a label for it is a process smell: it means someone stood at a printer while pot life burned. Write on the cup:

```
IN2 2142 JS          product · time mixed (HHMM, 24 h) · initials
XCR 0906 SS 140g     add batch mass where CS-004 §7.3 requires it
```

Four characters of time is what a gloved hand will actually write on a curved cup, and the minute is the only granularity that answers what a pot label is asked: is this still good, and whose is it. Anything unlabeled and curing is treated as hot waste (CS-008).

Anything opened but not given a lot record gets an opened sticker: `OPEN 2026-10-14 JS`.

7.8 When there is no label printer

This is the documented path, not a degraded one, and it stays the fallback after a printer is bought.

1. Print the label sheet on plain paper from the app.
2. Cut on the guillotine, on the crop marks. Not with scissors.
3. **Overlaminate with 3 in clear packing tape, overlapping the label by at least 6 mm on all four sides**, so the paper is fully encapsulated with no exposed edge anywhere. Burnish hard with a thumbnail.

   Done this way it survives repeated IPA wipes, CF dust and compressed air for months, because the solvent never reaches paper or toner. Done wrong, with the tape flush to the label edge, solvent wicks underneath and it delaminates in a week. **2 in tape over a 2 in label is exactly wrong.**

For anything tag-able, a laminating pouch on a zip tie beats packing tape, because the lamination seals all four edges. Punch the hole through the sealed margin, never through the paper.

7.9 When the printer breaks the week before comp

In order:

1. **The spare cartridge**, taped to the printer in a bag, permanently, never allowed to become the working roll. Most of "the label printer is broken" is an empty or jammed cartridge.
2. **Avery 5161 on the office laser** through the app, with the §7.8 overlaminate. Full fidelity, one grade down in durability, and no purchase decision at 11 pm. Keep one unopened box in the cabinet.
3. **Paint marker on masking tape.** Always works, never scannable, acceptable for 72 hours.

7.10 Storage locations

Every shelf, bin and rack holding composites material gets a storage record and a label on its front edge. That is what turns the location field into a controlled value rather than the freeform string CS-011 §7.3 flags, and it is what lets a move be recorded by scanning the object and then the shelf.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Every mold in shared space carries an engraved ID	100%	Monthly stock walk (CS-011)	Walk notes
Every mold carries a current printed label	100%	Monthly stock walk; a label whose seal date predates the last re-seal counts as failed	Walk notes
No unlabeled cloth in the cutting area	100%	Spot check at every layup	—
No label within 40 mm of a flange seal band	100%	Checked at bagging, before pulling vacuum	WO step note
Calibration bar measures 100 mm	Every print session	Steel rule	—
Coupon IDs written on the panel before the first cut	100%	Checked at the saw	Panel record
Parts carry their hidden-face label before clear coat	100%	CS-009 finishing check	WO quality checks
Hand marks use a paint marker, not a Sharpie	100%	Spot check	—

9. Records

The labels themselves, the records they point at, and the location field on each record, which is what the monthly stock walk checks against. A scan of any label resolves to the public page for that record, so a stock walk can be done with a phone and no account.